FLAT BASE CHANGE COMMUTES FLAT PULLBACK WITH PROPER PUSHFORWARD ON CYCLE CLASSES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a base change check for properness and flatness, a reduction to the structure sheaf of one integral closed subspace, and flat base change in cohomology

Editorial note. The label lemma-flat-pullback-proper-pushforward names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The cycle class of an integral closed subspace as the class of its structure sheaf, and the compatibility of flat pullback and proper pushforward with the cycle classes attached to coherent sheaves
- Stability of properness and of flatness of a given relative dimension under base change
- Flat base change in cohomology for a proper morphism, identifying the pushforward of the pulled back structure sheaf with the pullback of the pushforward

Source reference. Original reference 02QY: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 762–762. Prerequisite pointer invoked at proof line(s) 1526; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02R6: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1021–1021. Prerequisite pointer invoked at proof line(s) 1527; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02RE: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 1406–1406. Prerequisite pointer invoked at proof line(s) 1528; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof

Date: stacks-project:spaces-chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

line(s) 1499; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 073K: sources/stacks-chow/tags.txt, source lines 9064–9064. Prerequisite pointer invoked at proof line(s) 1532; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03MO: sources/stacks-chow/tags.txt, source lines 4756–4756. Prerequisite pointer invoked at proof line(s) 1522; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Prerequisite pointer invoked at proof line(s) 1518; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04NS: sources/stacks-chow/tags.txt, source lines 6090–6090. Prerequisite pointer invoked at proof line(s) 1521; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a fibre product diagram of good algebraic spaces over B. Assume $f : X \to Y$ proper and $g : Y' \to Y$ flat of relative dimension r. Then also f' is proper and g' is flat of relative dimension r. For any k-cycle α on X we have

$$g^* f_* \alpha = f'_* (g')^* \alpha$$

in $Z_{k+r}(Y')$.

Proof. The assertion that f' is proper follows from Morphisms of Spaces, Lemma 04WP. The assertion that g' is flat of relative dimension r follows from Morphisms of Spaces, Lemmas 04NS and 03MO. It suffices to prove the equality of cycles when $\alpha = [W]$ for some integral closed subspace $W \subset X$ of δ-dimension k. Note that in this case we have $\alpha = [\mathcal{O}_W]_k$, see Lemma 02QY. By Lemmas 02R6 and 02RE it therefore suffices to show that $f'_*(g')^* \mathcal{O}_W$ is isomorphic to $g^* f_* \mathcal{O}_W$. This follows from cohomology and base change, see Cohomology of Spaces, Lemma 073K. □